Chapter 2

Local Solvability of an Inverse Problem

We consider a theorem on the existence and a theorem of the uniqueness of a 'short-range' inverse problem solution that is for a body close to the given. We apply methods which are different from those for the three-dimensional case (Sretenskii, 1939; Ivanov, 1956b; Prilepko, 1967, 1970; Isakov, 1973, 1978); we use, in particular, a theory of the boundary-value problem for analytic functions. The proof of the existence of this theory is very constructive: successive approximations, allowing one to find the solution to a problem, are constructed in explicit form. Further we shall use it to construct a numerical algorithm of an inverse potential problem solution. We note two moments of principle in the investigation stated below. First, a linearized problem is completely investigated; second, difficulties involving a branching equation are resolved. The presentation is carried out by Cherednichenko (1976b). In Sections 10 and 11 the theorems on existence of an inverse problem solution for small constant densities are proved where the density estimate by the data of an inverse problem is constructed explicitly. The author's paper (Cherednichenko, 1982 e) presents the results and scheme of the proof. Section 9 observes the solvability of an inverse problem also for complex densities.

§2.1 Univalent function variation

Consider one simple assertion for univalent functions.

LEMMA 2.1. (variation of univalent functions) Let $z(t)$ be an analytic function for $|t| < 1$, continuous together with the first derivative and univalent for $|t| \leq 1$. Let

$$\min_{|t|=1} |z'(t)| \geq d > 0. \tag{2.1}$$

Then there is a number ε, $\varepsilon > 0$ such that for any function $\omega(t)$, being analytic for $|t| < 1$, continuous together with the first derivative for $|t| \leq 1$ and $|\omega'(t)| \leq \varepsilon$, $|t| = 1$ the function $z(t) + \omega(t)$ is univalent for $|t| \leq 1$.

PROOF. Introduce the function

$$f(t_0,t)=\begin{cases}\dfrac{z(t)-z(t_0)}{t-t_0}, & t\neq t_0\\[2ex] z'(t_0), & t=t_0,\ |t|\le 1,\ |t_0|\le 1.\end{cases}$$

Let $d_1=\min f(t,t_0)$ for $|t|\le 1$, $|t_0|\le 1$. Since $z(t)$ univalency and the condition (1) hold, we have $d_1>0$. To prove the lemma, it is required to show that $z(t)+\omega(t)-[z(t_0)+\omega(t_0)]$ vanishes only for $t=t_0$ or

$$\frac{z(t)+z(t_0)}{t-t_0}+\frac{\omega(t)-w(t_0)}{t-t_0} \tag{2.2}$$

does not vanish for $|t|\le 1$, $|t_0|\le 1$. We have

$$\left|\frac{\omega(t)-w(t_0)}{t-t_0}\right|=\frac{1}{|t-t_0|}\left|\int_{t_0}^{t}\omega'(\tau)d\tau\right|\le\max_{|t|\le 1}|\omega'(t)|\le\varepsilon.$$

If $\varepsilon<d_1$, then the representation in (2) does not vanish. Thus the lemma is proved.

§2.2 Local theorem of existence

1^0. Following Vekua (1988) introduce the function spaces.

Let $C(\overline{G})$ be a set of continuous functions in a closed domain $\overline{G}$ which are, generally speaking, complex-valued with the norm

$$C(f,\overline{G})=\max_{z\in\overline{G}}|f(z)|\,.$$

If f and its partial derivatives up to the order m are continuous in the closed domain $\overline{G}$, then we shall write $f\in C^m(\overline{G})$. It should be noted that, in a boundary point, the derivatives are determined as the limits of the derivatives of the same name in the domain. The norm f is determined by

$$C^m(f,\overline{G})\equiv C^m(f)=\sum_{k=0}^{m}\sum_{l=0}^{k}C\left(\frac{\partial^k f}{\partial x^{k-l}\partial y^l},\overline{G}\right).$$

Let us agree that $C^0=C$.

Let $f(z)$ on the closed domain $\overline{G}$ satisfy the condition

$$|f(z_1)-f(z_2)|\le H\,|z_1-z_2|^{\alpha}\,,\quad 0<\alpha\le 1,$$

where z_1,z_2 are arbitrary points from $\overline{G}$, and H, α are positive constants which do not depend on the z_1 and z_2 option. Denote the lower bound of the numbers H satisfying this inequality by $H(f)$ and call it the Hölder constant. It is obvious that

$$H(f)=H(f,\alpha,\overline{G})=\sup\frac{|f(z_1)-f(z_2)|}{|z_1-z_2|^{\alpha}},$$

$$|f(z_1) - f(z_2)| \leq H(f)\,|z_1 - z_2|^{\alpha}\,.$$

The last inequality is called the Hölder condition. Denote by $C_\alpha(\overline{G})$ the set of all bounded functions satisfying this inequality where the constant α, called the Hölder exponent of f, is the same for all elements of $C_\alpha(\overline{G})$. The norm f is obtained by the formula

$$C_\alpha(f, \overline{G}) = C_\alpha(f) = C(f, \overline{G}) + H(f, \alpha, \overline{G}).$$

The space $C^m_\alpha(\overline{G})$ has yet to be dealt with. The functions $C^m(B\gamma)$ are the elements of this space and satisfy the condition

$$\frac{\partial^m f}{\partial x^{m-k}\partial y^k} \in C_\alpha(\overline{G}), \quad k = 0, 1, \ldots, m.$$

The norm in this space is defined as follows

$$C^m_\alpha(f) = C^m(f) + \sum_{k=0}^{m} H(\frac{\partial^m f}{\partial x^{m-k}\partial y^k}, \alpha, \overline{G}).$$

2^0. We shall say that the Jordan curve S belongs to the class C^m_α, $m \geq 0$, $0 < \alpha \leq 1$, (C^m, $m \geq 1$) if there exists a parametric representation

$$z = x(\theta) + iy(\theta), \quad a \leq \theta \leq b,$$

where real functions $x(\theta)$, $y(\theta)$ belong to $C^m_\alpha[a, b]$ (C^m, $m \geq 1$) and $z'(\theta) \neq 0$, $a \leq \theta \leq b$. If S is a closed curve then $z(a) = z(b)$.

Let $z(t)$ conformally map the disk $|t| < 1$ onto a domain D^+ with a boundary S. The boundary properties of $z(t)$ and its inverse $t(z)$, evidently, depend on the smoothness of S. We shall use the following fact (Vekua, 1988): if $S \in C^m_\alpha$, $m \geq 0$, $0 < \alpha < 1$ then $z(t) \in C^m_\alpha(|t| \leq 1)$, $t(z) \in C^m_\alpha(\overline{D^+})$. It is clear that if $z(t) \in C^m_\alpha(|t| \leq 1)$, then, by the definition, $S \in C^m_\alpha$. Note that the condition $\alpha > 0$ is essential.

3^0. Now we shall formulate the basic result. Let D^+ be a domain with a boundary S of the class C^1_α, $0 < \alpha < 1$, and domains T_1^+, T_2^+ such that $\overline{T_1^+} \subset D^+$, $\overline{D^+} \subset T_2^+$. Let, further $\mu(z, \overline{z})$ be a real-valued function, $\mu \in C^1_\alpha(\overline{T_2^+})$ and

$$\mu(z, \overline{z}) \neq 0, \quad z \in S. \tag{2.3}$$

Then $\mu(z, \overline{z})$ is not zero in a certain neighborhood of S. In order not to introduce new denotations we shall consider $\mu(z, \overline{z}) \neq 0$ in $T_2^+ \backslash \overline{T_1^+}$. Denote by Ω the set in $\overline{D^-}$ which satisfies, at most, one of the properties:

a) Ω has a limit point distinct from $z = \infty$ in D^-;

b) the measure of $\Omega \bigcap S$ is positive.

THEOREM 2.1. Let in the domain T_1^- be given an analytic function $U(z)$, $U(\infty) = 0$, continuous in $\overline{T_1^-}$ and

$$|U(z)| \leq M_1, \quad z \in \overline{T_1^-}, \tag{2.4}$$

$$\lim_{|z|\to\infty} zU(z) = c_0, \quad \operatorname{Im} c_0 = 0. \tag{2.5}$$

There exists a positive number λ_* depending on D^+, M_1, μ, Ω such that if

$$\left|U^-(z) - U(z)\right| \le \lambda_*, \quad z \in \Omega, \tag{2.6}$$

holds, then such a domain D_*^+ with a boundary S_* of the class C_α^1 belonging to $T_2^+ \backslash \overline{T_1^+}$ that

$$U^-(z, D_*^+, \mu) = U(z), \quad z \in D_*^-$$

holds.

Note that condition (5) for the function $U(z)$ is necessary for the problem solvability because c_0 is the mass of the desired domain (see 4^0 of Section 1.1). This number (c_0) must be real because μ is real. From the proof it will follow that the domain D_*^+ is sought for near the given domain D^+.

§2.3 Linearization of the boundary value problem

1^0. We now turn to the proof of the theorem. Introduce the class of the domains

$$D_\omega^+ = \{z : \ z = z(t) + \omega(t), \ |t| < 1, \ \|\omega\| \le d_1\};$$

here $z(t)$ is a conformal mapping of the disk $|t| < 1$ onto D^+ with the condition that $z(0) = 0$ and $z'(0) > 0$ (assuming $0 \in T_1^+$); $\omega(t)$ is a function which is analytic for $|t| < 1$, $\omega \in C_\alpha^1(|t = 1|)$; $\|\cdot\|$ is a norm in this space; and

$$\omega(0) = 0, \quad \operatorname{Im} \omega'(0) = 0, \tag{2.7}$$

Positive d_1 is taken so that first, $\overline{T_1^+} \subset D_\omega^+$, $\overline{D_\omega^+} \subset T_2^+$, and, second, the function $z(t)+\omega(t)$ is univalent. The latter may be done due to lemma 2.1.

Given $\mu(z, \overline{z})$ we construct $M(z, \overline{z})$: $M_{\overline{z}}(z, \overline{z}) = \mu(z, \overline{z})$, $z \in T_2^+$ as was done in Section 1.2. It is known, that $M \in C_\alpha^2(\overline{T_2^+})$. Now, due to Theorem 1.1, the inverse problem is reduced to the nonlinear boundary-value problem: to find $\varphi_+(t)$, $\omega(t)$ analytic for $|t| < 1$, the limit values of which belong to $C_\alpha^1(|t| = 1)$, where

$$g^+(t) = M[z(t) + \omega(t), \overline{z(t) + \omega(t)}] - U[z(t) + \omega(t)], \quad |t| = 1. \tag{2.8}$$

Subtracting from (8) the equality

$$g_0^+(t) = M[z(t), \overline{z(t)}] - U^-[z(t)], \quad |t| = 1, \tag{2.9}$$

we extract the linear component and denoting $\psi_+ = g^+ - g_0^+$ we may write

$$\psi_+ = \mu[z, \overline{z}]\overline{\omega} + (M_z[z, \overline{z}] - (U^-)')\omega + B\omega, \quad |t| = 1, \tag{2.10}$$

$$\begin{aligned} B\omega = U^-[z] + (U^-)'[z]\omega - U[z + \omega] + M[z + \omega, \overline{z + \omega}] - \\ - M[z, \overline{z}] - M_z[z, \overline{z}]\omega - \mu[z, \overline{z}]\overline{\omega}, \end{aligned} \tag{2.11}$$

where $z = z(t)$, $\omega = \omega(t)$, $(U^-)'[z]$ is the derivative of $U^-(z)$ in the point $z = z(t)$ (due to our assumptions on the smoothness of S and $\mu(z,\overline{z})$ we have $U^-(z) \in C^2_\alpha(S)$).

2^0. Now we shall research the linear component of the boundary-value problem (10).

Let functions $a(t), b(t), c(t)$ satisfy the Hölder condition in the circle $|t| = 1$. The problem is to find the functions $\varphi_+(t), \varphi_-(t)$, analytic for $|t| < 1$ and $|t| > 1$ respectively, continuous in the closures of these domains by the boundary conditions

$$\varphi_+ = a\varphi_- + b\overline{\varphi_-} + f, \quad |t| = 1, \quad \varphi_-(\infty) = 0. \tag{2.12}$$

This is the generalized Hilbert problem, or the general linear conjugation problem. It is known that the problem is pertaining to Noether for $a(t) \neq 0$, and, in particular, to Fredholm, if the index $a(t)$ is zero. Recall, that the functional index $a(t)$, $a(t) \neq 0$, relative to the curve S, is the increment of the argument $a(t)$ divided by 2π. In the cases $|a| = |b|$ and $|b| < |a|$ the number of linearly independent solutions l over the real number field of the homogeneous problem is determined by the indexes of a, b (Mikhailov, 1963). In the other cases this problem is unsolved (Litvinchuk, 1977).

If we denote $\omega = \overline{\varphi_-}$, then (10) takes the form (12), where $a = \mu[z,\overline{z}]$, $b = M_z - (U^-)'[z]$ and, as it follows from the examples, neither of the two relations between a and b is fulfilled. Nevertheless we define l in this case. To this purpose we differentiate (9) by t :

$$(g_0^+)'(t) = \mu[z,\overline{z}]\frac{d\overline{z(t)}}{dt} + M_z[z,\overline{z}]z'(t) - U_z'[z]z'(t).$$

Taking into account $(\overline{z(t)})_t = -\overline{t^2 z'(t)}$ rewrite this equality

$$\frac{(g_0^+)(t)}{z'(t)} = -\overline{t^2}\frac{\overline{z'(t)}}{z'(t)}\mu + M_z[z,\overline{z}] - U_z'[z].$$

(Note, that these transformations take place in the case when $S \in C^1_\alpha$ and $\mu \in C^0_\alpha(S)$. We shall use this when proving Theorem 2.2.)

Now, (10) takes the form

$$\phi^+ = \mu[z,\overline{z}]\overline{\omega} + \overline{t^2}\frac{\overline{z'}}{z'}\mu[z,\overline{z}]\omega + B\omega, \tag{2.13}$$

where $\phi^+ = \psi_+ - (g_0^+)'\omega/z'$ is the new unknown function. If we find ϕ^+, ω from (13), then ψ_+ will also be determined.

The modulus of ω and $\overline{\omega}$ coefficients in (13) are equal. Using the results of Mikhailov (1963) we obtain that the homogeneous problem

$$\phi^+ = \mu[z,\overline{z}]\omega + \overline{t^2}\frac{\overline{z'}}{z'}\mu[z,\overline{z}]\omega, \quad \omega(0) = 0, \ |t| = 1, \tag{2.14}$$

has a unique nontrivial solution. Because this problem is pertaining to Fredholm (the index of $\mu[z,\overline{z}]$ is equal to zero) the inhomogeneous problem

$$\phi^+ = \mu[z,\overline{z}]\overline{\omega} + \overline{t^2}\frac{\overline{z'}}{z'}\mu[z,\overline{z}]\omega + f, \quad \omega(0) = 0 \tag{2.15}$$

is solvable if and only if the orthogonality on f holds.

It is clear that

$$\phi_0^+ = 0, \quad \omega_0 = itz'(t) \tag{2.16}$$

solves (14). The conjugated problem to this one is the problem of determining ψ_+, ψ_- by the boundary-value condition (Mikhailov, 1966; Vekua, 1970)

$$\psi_- = \mu[z,\overline{z}]\psi_+ - \frac{z'}{\overline{z'}}\mu[z,\overline{z}]\overline{\psi_+}, \quad \psi_-(\infty), \quad |t| = 1.$$

The unique solution of this problem is $\psi_+^0 = z'(t)$, $\psi_-^0 = 0$. From here also follows that (14) has a unique trivial solution. If is known (Mikhailov, 1966) that the problem is solvable if and only if

$$\operatorname{Re} \int\limits_{|t|=1} z'(t)f(t)dt = 0. \tag{2.17}$$

§2.4 The auxiliary problem

1^0. As follows from the example

$$z(t) = t, \quad U(z) = \frac{1}{z} + \frac{c_1}{z^2}, \quad \omega = a_2 t^2,$$

the condition

$$\operatorname{Re} \int\limits_{|t|=1} z'(t)B\omega dt = 0,$$

generally speaking, is false; therefore, we shall consider instead of (13) the other constant term

$$\phi^+ = \mu[z,\overline{z}]\overline{\omega} + \overline{t^2}\frac{\overline{z'}}{z'}\mu[z,\overline{z}]\omega + B_1\omega, \tag{2.18}$$

$$B_1\omega = B\omega + \frac{c(\omega)i}{tz'(t)}. \tag{2.19}$$

The functional $c(\omega)$ is constructed by $B\omega$ as follows

$$\operatorname{Re} \int\limits_{|t|=1} z'(t)B_1\omega dt = 0, \tag{2.20}$$

i.e.,

$$c(\omega) = -\frac{1}{2\pi} \operatorname{Re} \int\limits_{|t|=1} z'(t)B\omega dt.$$

Denote by $\tilde{C}_\alpha^1$ the set of functions $\omega(t)$ analytic for $|t| < 1$, $\omega \in C_\alpha^1(|t| = 1)$ and satisfying (7). Taking into account (16) we may conclude that in this space the homogeneous problem (14) has only a trivial solution; therefore, due to (20), the boundary-value problem (18) is equivalent to the equation

$$\omega = AB_1\omega, \tag{2.21}$$

where A is a linear bounded operator transforming functions $f \in C^1_\alpha(|t| = 1)$, which satisfy (17), into C^1_α. Further (7^0) we shall explicitly construct A.

2^0. In Section 2.5 we shall prove that (21) has a unique solution $\omega_*(t)$ in a certain ball $\|\omega\|_{\tilde{C}^1_\alpha} \le d_*$ for λ_* sufficiently small. Now we shall show that $\omega_*(t)$ solves (13). If ω_* solves (21), then (18) is solvable and, therefore,

$$\phi^+_* = M[z+\omega_*, \overline{z+\omega_*}] - U[z+\omega_*] + \frac{c(\omega_*)i}{tz'(t)}, \quad |t| = 1$$

holds. After multiplying this equality by $z' + \omega'_*$ and integrating we obtain

$$0 = \mathrm{Re}\Big[\int\limits_{|t|=1} M[z+\omega_*, \overline{z+\omega_*}](z'+\omega'_*)dt -$$

$$- \int\limits_{|t|=1} U[z+\omega_*](z'+\omega'_*)dt + c(\omega)i \int\limits_{|t|=1} \frac{z'+\omega'_*}{tz'}dr\Big],$$

or

$$0 = \mathrm{Re}\int\limits_{S_*} M[z,\overline{z}]dz - \mathrm{Re}\int\limits_{S_*} U[z]dz + c(\omega_*)\,\mathrm{Re}\left[-2\pi + i\int\limits_{|t|=1}\frac{\omega'_*}{tz'}dt\right], \tag{2.22}$$

where S_* is the boundary of $D^+_* = \{z : z = z(t) + \omega_*(t),\ |t| < 1\}$.

From the Borel-Pompeju (Green) formula follows that

$$\int\limits_{S_*} M[z,\overline{z}]dz = 2i\int\limits_{D^+_*} M_{\overline{z}}(z,\overline{z})dz = 2i\int\limits_{D^+_*} \mu(z,\overline{z})dxdy;$$

therefore, the first term in the right side of (22) vanishes, since $\mu(z,\overline{z})$ is a real function. The second term vanishes due to (5). Finally, if $\max_{|t|=1}|\omega'_*(t)| < \min_{|t|=1}|z'(t)|$, then

$$\mathrm{Re}\int\limits_{|t|=1}\frac{i\omega'_*}{tz'}dt \le \left|\int\limits_{|t|=1}\frac{i\omega'_*}{tz'}dt\right| \le 2\pi\max_{|t|=1}\left|\frac{w'_*}{z'}\right| < 2\pi.$$

So, from (22) follows $c(\omega_*) = 0$, i.e., (8) holds, and, therefore, D^+_* solves the inverse problem.

§2.5 The Newton-Kantorovitch method.

1^0. Here we shall prove (21) solvability using the scheme of the Newton-Kantorovitch method (Kantorovich and Akilov, 1977). We shall show that in the ball $\|\omega\| \le d \le d_1$, where d_1 is from the D^+_ω definition and d is unknown, the following estimate

$$\|AB_1\omega\| \le a + bd + cd^2, \tag{2.23}$$

holds, where a, b may be done arbitrarily small due to the choice of λ_* in (6). Equating the right side of (23) to d we obtain the quadratic equation, the lesser root of which is

$$d_* = \left((1-b) - \sqrt{(1-b)^2 - 4ac}\right)/2a.$$

So, $AB_1\omega$ maps the ball $\|\omega\| \le d_*$ into itself. Verifying the contractibility of this operator we obtain the inequality

$$\|A\omega_1 - A\omega_2\| \le (b + 2ad_*)\,\|\omega_1 - \omega_2\|.$$

On the basis of the definition of d_*, therefore, we have $b + 2ad_* < 1$. Hence, (21) has a unique solution in the ball $\|\omega\| \le d_*$.

Now we shall install (23). First, from (19), (20) we have

$$\|B_1\omega\| \le l_1\,|A|\,\|B\omega\|,$$

where l_1 depends only on $z(t)$. $|A|$ is the norm of A. Further, let $B_1\omega = B_2\omega + B_3\omega$, where

$$B_2\omega = U^-[z] - U[z] + \left(U^-[z] - U[z]\right)'\omega - [U[z+\omega] - U[z] - U'[z]\omega],$$

$$B_3\omega = M[z+\omega, \overline{z+\omega}] - M[z,\overline{z}] - M_z[z,\overline{z}]\omega - \mu[z,\overline{z}]\omega.$$

From here we obtain

$$\|B_2\omega\| \le l_2\left\|U^- - U\right\|_{C^1_\alpha(S)} + dl_3\left\|U^- - U\right\|_{C^1_\alpha(S)} + \frac{M_2}{2}d^2,$$

$$\|B_3\omega\| \le \frac{M_3}{2}d^2.$$

Here l_2, l_3 depend on $z(t)$, M_2, M_3 and on $\max|U''(z)|$ in $\overline{T^-}$, $\max(|\mu_x|, |\mu_y|)$ in $\overline{T_2^+}\backslash T_1^+$. Combining these estimates we obtain (23), where a, b tend to zero for $\lambda \to 0$ and

$$\left\|U^- - U\right\|_{C^2_\alpha(S)} \le \lambda. \tag{2.24}$$

2^0. It remains to prove that for each $\lambda > 0$ there exists such λ_*, that from (6) follows (24). Assume the contrary (Isakov, 1980). Then there exists such $\varepsilon > 0$ and $U_n(z)$ analytic in T_1^-, continuous in $\overline{T_1^-}$, satisfying (4) that

$$\left|U^- - U_n\right| \le \frac{1}{n}, \quad z \in \Omega,$$

But

$$\left\|U_1^- - U_n\right\|_{C^2_\alpha}(S) \ge \varepsilon. \tag{2.25}$$

Since $|U_n| \le M_1$ in $\overline{T_1^-}$, then, due to the Cauchy inequality,

$$\left|U_n^{(3)}(z)\right| \le M_4, \quad z \in \overline{D^-}$$

holds, where M_4 depends on M_1. Hence follows, that from U_n we may choose the converging in $C^2_\alpha(\overline{D_3^-})$ subsequence U_{n_p}. The limit element we denote by U_∞. By construction we have $U_\infty(z) = U^-(z)$, $z \in \Omega$. From here, by the uniqueness theorem for analytic functions, we have $U_\infty(z) = U^-(z)$, $z \in D^-$. Therefore, $\left\|U_{n_p} - U^-\right\|_{C^2_\alpha(S)}$ tends to zero for $n_p \to \infty$ which contradicts (25).

§2.6 The explicit solution of the linear problem.

1^0. The solution $\omega_*(t)$ of (21) is the limit of the successive approximations

$$\omega_{n+1} = AB_1\omega_n, \quad n = 0, 1, 2, \ldots,$$

where ω_0 is an arbitrary element from $\|\omega\| \leq d_*$.

We shall construct A explicitly. To this purpose we solve boundary-value problem (18). It may be written as

$$F^+ = tz'\mu[z,\overline{z}]\overline{\omega} + \overline{tz'}\mu[z,\overline{z}]\omega + tz'B_1\omega, \quad |t| = 1, \tag{2.26}$$

where $F^+ = tz'\Phi^+$. Hence we obtain for F^+ the following boundary condition

$$F^+ - \overline{F^+} = tz'B_1\omega - \overline{tz'B_1\omega}, \quad F^+(0) = 0.$$

Solution to this problem is given by the Schwartz formula

$$F^+(t) = \frac{1}{2\pi i}\int\limits_{|\tau|=1} \frac{\tau z'(\tau)B_1\omega - \overline{\tau z'(\tau)B_1\omega}}{\tau - t}d\tau, \quad |t| < 1.$$

Boundary condition (26) takes a form

$$\operatorname{Re}\left[\overline{z'}\mu[z,\overline{z}]\frac{\omega(t)}{t}\right] = \frac{1}{2}\left[F^+ - tz'B_1\omega\right], \quad |t| = 1. \tag{2.27}$$

Here $F^+(t)$ is a known function. The problem of determining $\omega(t)$, analytic for $|t| < 1$ by the condition (27) is called the Riemann-Hilbert problem. It is easy to show that $iz'(t)$ solves homogeneous problem (27), and in this case the solution of the inhomogeneous problem is the following

$$\omega(t) = t\left[\frac{z'}{2\pi i}\int\limits_{|\tau|=1} \frac{F^+(\tau) - \tau z'B_1\omega}{|z'|^2\,\mu(\tau - t)}d\tau - \frac{z'}{4\pi i}\int\limits_{|\tau|=1} \frac{F^+(\tau) - \tau z'B_1\omega}{|z'|^2\,\mu\tau}d\tau\right].$$

This formula shows us that the successive approximations ω_{n+1} may be obtained explicitly.

So, we have proved the existence of the 'short-range' inverse potential problem solution and we have given its construction. The uniqueness of the 'short-range' inverse potential problem solution exists and for the weaker restrictions.

§2.7 The local uniqueness theorem.

1^0.

THEOREM 2.2. Let D^+ be a domain with a boundary S of the class C^1_α, $\mu \in C^0_\alpha(\overline{T_2^+})$, $\overline{T_2^+} \supset \overline{D^+}$ and D_1^+ be a domain with a boundary of the class C^0_α. There exists such

a number d_{**} depending on D^+ and μ, that if $\|z(t) - w(t)\| \leq d_{**}$, where $z(t)$, $w(t)$ conformally map the disk $|t| < 1$ onto D^+ and D_1^+ correspondingly with conditions $z(0) = w(0) = 0$, $z'(0) > 0$, $w'(0) > 0$, then from the equality

$$U^-(z, D^+, \mu) = U^-(z, D_1^+, \mu), \quad z \in D^-$$

follows $D^+ = D_1^+$.

PROOF. We seek for a function $\omega_*(t)$ in the space $C_\alpha^1(|t| = 1)$ and not in $C_\alpha^0(|t| = 1)$ only to guarantee the function $z(t) + \omega_*(t)$ to be univalent in the disk $|t| \leq 1$. Equation (21) has a unique solution in the ball $\|\omega\|_{C_\alpha^0} \leq d_{**}$ for sufficiently small d_{**}, but (21) for $U(z) = U^-(z, D^+, \mu)$ has, evidently, the trivial solution $\omega = 0$. Hence follows Theorem 2.2.

2^0. Now we shall construct the non-uniqueness example of the 'short-range' inverse problem solution in the case of the variable domain. We shall show that for each $\varepsilon > 0$ there exist such simply connected domains D^+ and T^+, $D^+ \neq T^+$ ($D^+ \bigcup T^+$ is also a domain) and the density $\mu(z, \overline{z})$ given in $D^+ \bigcup T^+$ that

$$U^-(z, D^+, \mu) = U^-(z, T^+, \mu), \quad \rho(D^+, T^+) < \varepsilon.$$

Here by $\rho(D^+, T^+)$ is denoted the area of the set $(D^+ \bigcup T^+) \backslash (D^+ \bigcap T^+)$.

Let B_r be the disk $|z| < r$, $\mu_r(x, y) = x^2 + y^2 - r^2/2$ and D_r^+ be a simply connected domain such that

a) D_r^+ has a common part of the boundary with B_r;

b) the set $D_r^+ \bigcup B_r = T_r^+$ is the domain with a Jordan boundary;

c) the area of D_r^+ for all r is greater than a certain number.

Evidently, such a sequence of domains may be constructed. Due to c) it is not constructed to the point and $\rho(D_r^+, T_r^+) = \pi r^2$. As it is easy to show $U^-(B_r, z) \equiv 0$, therefore,

$$U^-(T_r^+, z) = U^-(D_r^+ \bigcup B_r, z) = U^-(D_r^+, z),$$

what was to be proved.

Note, that one of the domains D_r^+, T_r^+ we may construct with the boundary of class C_α^m, $m > 1$, but the boundary of the other domain will not be smooth. Below we present two examples.

§2.8 The density variation. The equivalent solution set of the inverse problems. Remarks.

1^0. The following statements follow from Theorem 2.1 and are most relevant in magnetic prospecting (Gravity Prospecting, 1981, 1990).

THEOREM 2.3. Let D^+ be a domain with a regular analytic boundary S, μ be real analytic in T_2^+, $T_2^+ \supset \overline{D^+}$ function, $\mu(z, \overline{z}) \neq 0$, $z \in S$. There exists such a number $\varepsilon > 0$ depending on D^+, μ, that for any analytic function $v(z, \overline{z})$ such that

$$\max_{z \in \overline{T_2^+}} |\mu(z, \overline{z}) - v(z, \overline{z})| < \varepsilon$$

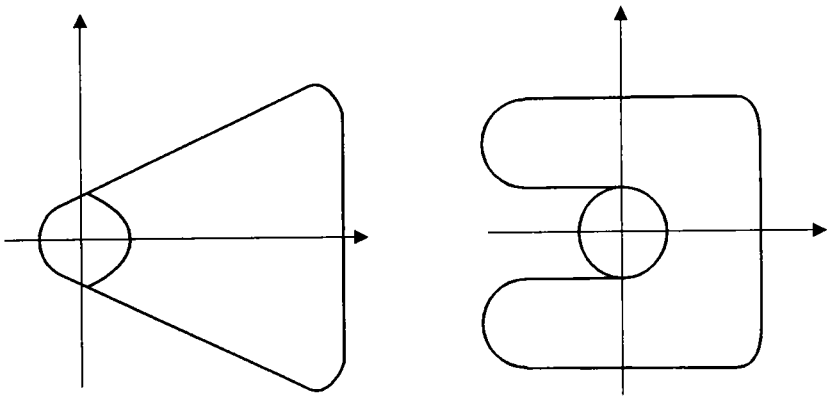

Figure 5: Local non-uniqueness example.

there exists a unique domain D_v^+, where

$$U^-(z, D^+, \mu) = U^-(z, D_v^+, v), \quad z \in D^-. \tag{2.28}$$

Really, if we denote $U(z) = U^-(z, D^+, \mu)$, then, due to the analyticity of S and μ, the function $U(z)$ has an analytic continuation from D^- into a certain domain T_1^-, $D^+ \supset \overline{T_1^+}$ (see Section 1.3) and

$$\left|U^-(z, D^+, v) - U(z)\right| = \frac{1}{\pi}\left|\int\limits_{D^+} \frac{\mu - v}{\zeta - z} d\xi d\eta\right| \leq \varepsilon k, \quad z \in \Omega,$$

where Ω may be, for example, a bounded line in D^-, k depends only on D^+ and Ω. We shall solve the inverse problem 'near' the domain D^+ with the density v. If ε is sufficiently small, then, due to Theorem 2.1 there exists a unique domain D_v^+, for which (28) holds.

The set of pairs (D_v^+, v) is called the equivalent set, because the domains D_v^+ with the density v give the same external potential. The case of the constant densities v will be considered further more explicitly.

2^0. Now we shall give some remarks.

REMARK 2.1. In comparison with the results of Prilepko (1966b), Isakov (1973) on the local solvability of the inverse potential problem in the three-dimensional case, in Theorem 2.1 it requires less smoothness of the density μ, namely $\mu \in C_\alpha^1$ (in Isakov (1973), for example, $\mu \in C_\alpha^4$).

REMARK 2.2. From the results on the smoothness of inverse problem solutions stated below (Section 6.1) it follows that under the conditions of Theorem 2.1 the boundary S_* of the desired domain belongs to C_α^2.

REMARK 2.3. The condition of boundary D^+ smoothness (regularity) where we determine the inverse potential problem is critical for the validity of Theorem 2.1. So, in the

example from Section 1.4 we have

$$z(t) = 3\left(t + \frac{1}{2}t^2\right), \quad \mu = 1, \quad U^-(z) = \frac{1}{z} + \frac{1}{z^2}.$$

For the function $U(z) = 1/z + c_1/z^2$ we have no solution for $c_1 > 1$. This is so because S in this case is analytic but not a regular curve, $z'(-1) = 0$.

REMARK 2.4. The uniqueness of the local solution to the inverse logarithmic potential problem in the case of a constant density was proved by V.K. Ivanov in his doctoral dissertation. The case of analytic density was researched by Prilepko (1974). This problem was also considered by Strakhov (1972), Pavlov (1975), Prilepko (1985).

REMARK 2.5. The method of local solvability research considered here was used by Suljandziga (1976) in the more general case, for solving the inverse problem of summarized potentials of masses and layers.

REMARK 2.6. The problem on the existence of the equivalent solutions set near the domain with the piecewise analytic boundary, for example, near the polygon remains open.

REMARK 2.7. The numerical construction of the equivalent solutions set of the inverse problem based on this method, is given in Section 9.3.

§2.9 The complex-valued density case.

We shall prove the theorem on the local solvability of the inverse problem for the complex density. This result arises from cooperation with G.V. Verjovkina.

1^0. At first we shall prove an auxiliary fact.

LEMMA 2.2. Let D^+ be a domain with a Jordan rectifiable boundary S, $\mu(z) \in A(D^+)$; then

$$\frac{1}{2i}\int_S \mu(z)U^-(z, D^+, \overline{\mu(z)})dz = \int_{D^+} |\mu(z)|^2\, dxdy \tag{2.29}$$

PROOF. Given $\mu(z)$, we construct the function

$$M(z) = \int \mu(z)dz.$$

Evidently, $M \in C^1(\overline{D^+})$. On S we obtain for $U^-(z) = U^-(z, D^+, \overline{\mu(z)})$ the following $\varphi_+ = \overline{M} - U^-$. Multiply both sides of this equation by $\mu(z)$ and integrate along S. The integral from the left is equal to zero. Use the Green formula

$$\int_{D^+} W_{\bar z} dxdy = \frac{1}{2i}\int_S W(z)dz.$$

This gives

$$\int_S \mu(z)U^-(z)dz = \int_S \mu(z)\overline{M(z)}dz = 2i\int_{D^+} \left(\mu(z)\overline{M(z)}\right)_{\bar z} dxdy =$$

$$= 2i \int\limits_{D^+} \mu(z)\overline{\mu(z)}dxdy = 2i \int\limits_{D^+} |\mu(z)|^2 \, dxdy.$$

Note that the loop S in (29) may be replaced by any loop S_1, which is contained in the analyticity domain of the function $\mu \cdot U^-$. If, for example, $\mu(z)$ is an entire function, then S_1 may be any loop which contains D^+ inside. Formula (29) yields

$$\frac{1}{2i} \int\limits_S \mu(z)U^-(z)dz > 0. \tag{2.30}$$

If μ is a real function, $\mu \geq 0$, then the necessary condition of the inverse problem solvability for $U(z)$ with the density μ is the following

$$\lim_{x\to\infty} zU(z) = c_0 > 0.$$

The constant c_0 may be found from

$$c_0 = \frac{1}{2\pi i} \int\limits_{S_1} U(z)dz,$$

where S_1, is an arbitrary closed loop containing D^+ inside, or, more precisely, containing the singularities of $U(z)$. Now, the inequality

$$\frac{1}{2i} \int\limits_{S_1} \mu(z)U(z)dz > 0, \tag{2.31}$$

where S_1 is any loop lying in the domain of analyticity of $\mu(z)U(z)$ is the necessary solvability condition of the inverse problem for $U(z)$ with the density $\overline{\mu(z)}$. In some cases we can avoid integrating. Let, for example, $\mu(z)$ be a polynomial and $U(z)$ be of the following form in a neighborhood of $z = 0$

$$\mu(z) = \sum_{k=0}^{n} a_k z^k, \quad U(z) = \sum_{k=1}^{\infty} \frac{c_{k-1}}{z^k}.$$

Then (31) takes the form

$$\pi(a_0c_0 + a_1c_1 + \ldots + a_nc_n) > 0.$$

If this condition is false, then the inverse problem has no solution.

Note, that the densities of the form $\overline{\mu(z)}$, $\mu \in A(D^+)$ have certain important properties. For example, in Section 7.3 it will be proved that, for each density $\mu(z, \overline{z})$, there exists a unique density $\overline{\mu_0(z)}$, where

$$U^-(z, D^+, \mu(z, \overline{z})) = U^-(z, D^+, \overline{\mu_0(z)}), \quad z \in D^-.$$

2^0. Now we shall formulate the analog of Theorem 2.1 for densities of the form $\overline{\mu(z)}$.

Let D^+ be a domain with boundary $S \in C^2_\alpha$, $0 < \alpha < 1$, T_1^+, T_2^+ be auxiliary domains $\overline{T_1^+} \subset D^+ \subset \overline{D^+} \subset T_2^+$. Let, further, $\mu \in A(T_2^+)$, $\mu(z) \neq 0$, $z \in T_2^+$.

THEOREM 2.4. Let in T_1^- be given an analytic function $U(z)$, $U(\infty) = 0$, and (31) hold, where S_1 is an arbitrary loop in $T_2^+ \backslash \overline{T_1^+}$ containing T_1^+. There exists such positive λ_*, depending on D^+, M_1, given by (4), μ, Ω, that, if

$$\left|U^-(z, D^+, \overline{\mu(z)}) - U(z)\right| \leq \lambda_*, \quad z \in \Omega.$$

Then a domain D_*^+ with a regular analytic boundary S_* situated in $T_2^+ \backslash \overline{T_1^+}$ exists such that

$$U^-(z, D_*^+, \overline{\mu(z)}) = U(z), \quad z \in D_*^-.$$

PROOF. By a method similar to that in Theorem 2.1, instead of (13) we obtain

$$\phi^+ = \overline{\mu[z]}\overline{\omega} + \overline{t^2}\frac{\overline{z'}}{z'}\overline{\mu[z]}\omega + B\omega. \tag{2.32}$$

Note, that the index of $\overline{\mu(z)}$ relative to S is equal to zero, because $\mu(z) \neq 0$ in T_2^+. The corresponding homogeneous problem, as earlier, has a unique solution $\phi^+ = 0$, $\omega = itz'(t)$. Conjugated to the homogeneous problem is the problem (Litvinchuk, 1977; Mikhailov, 1963) of determining the functions ψ_+, ψ_+ by the boundary condition

$$\psi_- = \overline{\mu[z(t)]}\psi_+ - \frac{z'(t)}{\overline{z'(t)}}\overline{\mu[z(t)]}\overline{\psi_+}, \quad \psi_-(\infty) = 0,$$

and $\psi_+ = z'(t)\mu[z(t)]$, $\psi_- = 0$ solve this problem. Therefore, to solve (32), it is necessary and sufficient that

$$\operatorname{Re} \int\limits_{|t|=1} z'(t)\mu[z(t)]B\omega dt = 0$$

holds.

Instead of (32) we consider the boundary-value problem

$$\phi^+(t) = \overline{\mu[z]}\overline{\omega} + \overline{t^2}\frac{\overline{z'}}{z'}\overline{\mu[z]}\omega + B_1\omega$$

$$B_1\omega = B\omega + \frac{c(\omega)i}{tz'(t)\mu[z(t)]}. \tag{2.33}$$

The functional $c(\omega)$ is constructed by $B\omega$ from the condition

$$\operatorname{Re} \int\limits_{|t|=1} z'(t)\mu[z(t)]B_1\omega dt = 0,$$

$$c(\omega) = -\frac{1}{2\pi i}\operatorname{Re} \int\limits_{|t|=1} z'(t)\mu[z(t)]B\omega dt. \tag{2.34}$$

Let ω_*, $\psi^+(t)$ solve (33), then

$$\psi^+(t) = \overline{M[z(t) + \omega_*(t)]} - U[z(t) + \omega_*(t)] + \frac{c(\omega_*)i}{tz'(t)\mu[z(t)]}.$$

Multiply this equality by $\mu[z+\omega_*][z'+\omega'_*]$ and integrate it along the unit circle:

$$0 = \operatorname{Re}\left[\int\limits_{|t|=1} \mu[z+\omega_*]M[z+\omega_*](z'+\omega_*)dt - \right.$$

$$\left. - \int\limits_{|t|=1} U[z+\omega_*]\mu[z+\omega_*](z'+\omega_*)dt + c(\omega_*)i \int\limits_{|t|=1} \frac{\mu[z+\omega_*](z'+\omega'_*)}{tz'(t)\mu[z(t)]}dt\right]. \tag{2.35}$$

Similarly as we prove the Lemma 2.2, we may show that the first integral in the square brackets is equal to

$$2i \int\limits_{D_*^+} |\mu(z)|^2\, dxdy. \tag{2.36}$$

The latter integral in (35) may be transformed as follows

$$\int\limits_{|t|=1} \frac{z'+\omega'_*}{tz'}dt + \int\limits_{|t|=1} \frac{\mu[z+\omega_*]-\mu[z](z'+\omega')}{tz'(t)\mu[z(t)]}dt.$$

Here the second integral is equal to zero, because $\mu[z(0)+\omega_*(0)] = \mu[z(0)] = \mu(0)$. Now (35) takes a form

$$0 = -\operatorname{Re}\int\limits_{S_*} U(z)\mu(z)dz + c(\omega_*)\operatorname{Re}\left(i\int\limits_{|t|=1} \frac{z'+\omega'_*}{tz'}dt\right). \tag{2.37}$$

Condition (31) denotes that

$$\operatorname{Im}\frac{1}{2i}\int\limits_{S_1} U(z)\mu(z)dz = 0 \tag{2.38}$$

or

$$\operatorname{Re}\int\limits_{S_1} U(z)\mu(z)dz = 0. \tag{2.39}$$

Due to the analyticity of $U(z)\mu(z)$ in $T_2^+\backslash\overline{T_1^+}$, the loop S_1 in the latter integral may be substituted by S_*, so

$$0 = c(\omega_*)\operatorname{Re}\left(i\int\limits_{|t|=1} \frac{z'+\omega'_*}{tz'}dt\right). \tag{2.40}$$

As was proved in Section 2.4, the factor before $c(\omega_*)$ is not vanishing; therefore, $c(\omega_*) = 0$, i.e., D_*^+ solves the inverse problem. The proof is complete.

§2.10 Existence theorems for the inverse problem for small constant densities

In this and in the next sections we shall obtain the sufficient conditions for inverse problem solvability. This conditions will be in the form of restrictions on the density of the desired domain. The existence of such conditions follows from the Theorem 2.1 on the local solution of the problem which states that if the given $U(z)$ is sufficiently close to $U^-(z)$, which is the potential of a certain domain D^+ with a density μ, then for $U(z)$ there exists a solution. The result has a conditional character, because, *a priori*, we do not know if there exists such a domain D^+ and a density μ for a given $U(z)$. Here we shall induce the inequality $\mu < \mu_*$, where μ_* may be found explicitly by $U(z)$, which guarantee the existence of the inverse problem solution with a constant positive density μ. This statement, being local, is unconditional.

The proof is based on the fact that $U(z)$ on the circle $|z| = r$, for r sufficiently large, is close to the function c_0/z, which is the potential of this disk with the density μ, $\mu r^2 = c_0$. Making more concrete the proof of the local existence theorem we obtain the sufficient solvability conditions of the inverse problem. The principal moment is that the research is carried out in the continuous functions space, but not in C^m_α, as in Theorem 2.1, because of the difficulties of calculating the norms of certain functions and operators in C^m_α. To this purpose we shall use the special method, based on the analytic continuation of the inverse problem in this case.

The results stated here are published by Cherednichenko (1981).

1^0. Let $U(z)$ be an analytic function for $|z| \geq R$ of the form

$$U(z) = \frac{c_0}{z} + \frac{c_2}{z^3} + \frac{c_3}{z^4} + \dots, \quad c_0 > 0. \tag{2.41}$$

We shall solve the inverse problem for $U(z)$ with the positive constant density μ. In (41) there is no term with $1/z^2$, but we may reduce the general case to this one as follows

$$\frac{c_0}{\xi} + \frac{c_1'}{\xi^2} + \dots = \frac{c_0}{(\xi - \xi_0) + \xi_0} + \frac{c_1'}{(\xi - \xi_0 + \xi_0)^2} + \dots =$$

$$= \frac{c_0}{\xi - \xi_0}\left(1 + \frac{\xi_0}{\xi - \xi_0}\right)^{-1} + \frac{c_1'}{(\xi - \xi_0)^2}\left(1 + \frac{\xi_0}{\xi - \xi_0}\right)^{-2} + \dots =$$

$$= \frac{c_0}{\xi - \xi_0} + \frac{c_1' - \xi_0 c_0}{(\xi - \xi_0)^2} + \dots$$

Assuming $\xi_0 = c_1'/c_0$ and introducing the new variable $z = \xi - \xi_0$ we obtain (41). This transformation may map a polynomial in $1/z$ into the function with an infinite number of terms and *vice versa*. The condition $c_1 = 0$ in (41) denotes that the center of gravity of the desired domain D^+ is situated in the point $z = 0$.

To compare the function $U(z)$ with the function $U_1 = c_0/z$ we introduce the numbers $\lambda, \lambda_1, \lambda_2$:

$$|U(z) - U_1(z)| \leq \lambda, \quad |z| = R, \tag{2.42}$$

$$|(U(z) - U_1(z))'| \leq \lambda_1, \quad |(U(z) - U_1(z))''| \leq \lambda_2, \quad |z| = R. \tag{2.43}$$

The main results are as follows.

THEOREM 2.5. Let a function $U(z)$ of the form (41) be analytic for $|z| \geq R$ and in $|z| = R$ inequalities (42), (43) hold. Then, for $\mu \leq \mu_*$,

$$\mu_* = \frac{c_0}{r_*^2}, \quad r_* = \max\left(\frac{5}{3}R, \sqrt{\frac{x_1}{c_0}}\right),$$

$$x_1 = 15\lambda R^3 + 9\lambda_1 R^4 + R^3(225\lambda^2 + 270\lambda\lambda_1 R + 48\lambda\lambda_2 R^2)^{1/2}$$

the inverse potential problem has a solution.

THEOREM 2.6. Let a function $U(z)$ of the form (41) be analytic for $|z| \geq R$ and for $|z| = R$ inequality (42) holds. Then, for $\mu \leq \mu_*$,

$$\mu_* = \frac{c_0}{r_*^2}, \quad r_* = \max\left(3R, \sqrt{\frac{x_1}{c_0}}\right), \quad x_1 = 130\lambda R^3$$

the inverse potential problem has a solution.

2^0. Here we shall give some remarks and examples relevant to these theorems.

REMARK 2.8. Theorem 2.6 gives a more approximate result than Theorem 2.5 but, in return, we need not estimate first and second derivatives of function $U(z) - U_1(z)$ for $|z| = R$.

REMARK 2.9. The number R is, in essence, the parameter of the problem. For one and the same $U(z)$ we may take various R and apply the theorems with $\lambda(R)$, $\lambda_1(R)$, $\lambda_2(R)$. From the obtained $\mu_*(R)$ we have to single out the greatest.

REMARK 2.10. Domain D_*^+ solving the inverse problem for μ_* may be constructed by the successive approximations method which will be described when proving theorems.

REMARK 2.11. The reiterated application of theorems using the obtained information on μ_* leads to the revision of μ_* estimates. This and the other refinements of theorems will be considered in more detail in 10^0, Section 2.11.

EXAMPLE 2.1. Let

$$U(z) = \frac{1}{z} + \frac{1}{z^3},$$

then, for $R = 1$ we have $\lambda = 1$, $\lambda_1 = 3$, $\lambda_2 = 12$. Theorem 2.5 gives $r_* = 8.8435$, $\mu_* = 0.0128$ and Theorem 2.6 gives $r_* = 11.4017$, $\mu_* = 0.0077$. To compare, note that from the necessary conditions stipulated in Section 4.7 follows the unsolvability of the problem for $\mu > 0.1875$ for the given function $U(z)$.

EXAMPLE 2.2. Let

$$U(z) = \frac{1}{z} + \frac{1}{z^3} + \frac{1}{z^4},$$

then for $R = 1$, we have $\lambda = 2$, $\lambda_1 = 7$, $\lambda_2 = 32$. On the base of Theorem 2.5 we obtain $r_* = 13.4553$. If we use Theorem 2.5 for $R = 4$, it gives $r_* = 10.3401$, and for $R = 6$, $r_* = 10$, $\mu_* = 0,01$. This problem is unsolvable for $\mu > 0.1008$ (it follows from Section 4.7 results). In this example we see the influence of R.

EXAMPLE 2.3. Let

$$U(z) = \frac{1}{z}\,\mathrm{e}^{1/z^2}.$$

Here $c_0 = 1$. Further,

$$|U(z) - U_1(z)| = \left|\frac{1}{z}(\mathrm{e}^{1/z^2} - 1)\right| = \left|\frac{1}{z}\sum_{k=1}^{\infty}\frac{1}{k!z^{2k}}\right| \le$$

$$\le \frac{1}{|z|}\sum_{k=1}^{\infty}\frac{1}{k!}\frac{1}{|z|^{2k}} = \frac{1}{|z|}\left(\mathrm{e}^{1/|z|^2} - 1\right).$$

From Theorem 2.6 for $R = 3.7$ it follows that $r_* = 11.6132$, $\mu_* = 0.0074$.

§2.11 Proof of theorems

1^0. Here we shall prove Theorem 2.5. The scheme will be the same as in Theorem 2.1. Solution of the inverse problem, domain D^+, we shall search 'near' the circle $|z| = r$; r is an as-yet unknown parameter. Therefore, the conformal map of the disk $|t| < 1$ onto D^+ has a form $rt + \omega(t)$, where $\omega(t)$ is analytic in $|t| < 1$ and

$$\omega(0) = 0, \quad \operatorname{Im}\omega'(0) = 0. \tag{2.44}$$

Now, due to Theorem 2.1, the inverse problem is equivalent to determining the functions $\varphi_+(t)$, $\omega(t)$ from the set $A(|t| < 1)$, i.e., analytic for $|t| < 1$ and continuous for $|t| \le 1$, by the nonlinear boundary condition

$$\varphi_+(t) = \mu\overline{(rt + \omega(t))} - U[rt + \omega(t)], \quad |t| = 1. \tag{2.45}$$

Now we shall single out the linear part in (45). The function $U_1(z) = c_0/z$ is the potential of the disk $|t| < r$ with the density $\mu = c_0/r^2$; therefore,

$$0 = \mu\overline{rt} - U_1^-(rt), \quad |t| = 1.$$

By means of this identity we shall write (45) in the form

$$\varphi_+(t) = \mu\overline{\omega} - U_1'(z)\omega + U_1(z) + U_1'(z)\omega - U[z + \omega], \quad z = rt, \tag{2.46}$$

where $U_1'(z)$ is the derivative of $U_1(z)$ with respect to z for $z = rt$

$$U_1'(z) = -\frac{c_0}{z^2} = -\frac{\mu}{t^2}.$$

Now (46) takes a form

$$\varphi_+ = \mu\overline{\omega} + \mu\frac{1}{t^2}\omega + U_1(z) - U(z) + B\omega, \quad |t| = 1, \tag{2.47}$$

$$B\omega = (U_1(z) - U(z))'\omega - (U(z + \omega) - U(z) - U'(z)\omega). \tag{2.48}$$

2^0. We shall solve the boundary-value problem (47) considering it as the linear one with the constant term $U_1 - U + B\omega$. To this purpose we shall, instead of (47), solve the auxiliary problem

$$\varphi_+ = \mu\overline{\omega} + \mu\frac{\omega}{t^2} + U_1(z) - U(z) + B_1\omega, \quad |t| = 1,$$

$$B_1\omega = B\omega + \frac{c(\omega)i}{tr}, \quad c(\omega) = \frac{1}{2\pi}\operatorname{Re}\int\limits_{|t|=1} rB\omega dt, \tag{2.49}$$

which in contrast to (47) is unconditionally solvable. Now, if we denote $\omega'(0) = \alpha$, α be real due to (44), and introduce a new unknown function $\psi_+ = \varphi_+ - \mu\,(\omega/t^2 - \alpha/t)$, then boundary condition (49) will be equivalent to the following

$$\psi_+ = \mu\overline{\omega} + \mu\frac{\alpha}{t} + U_1(z) - U(z) + B_1\omega, \quad |t| = 1. \tag{2.50}$$

Integrating both parts of (50) and taking into account that $U_1 - U$ is analytic for $|t| \geq 1$ and has zero of the third order for $t = \infty$, we obtain

$$0 = 2\mu\alpha + \frac{1}{2\pi i}\int\limits_{|t|=1} B_1\omega dt.$$

On the base of this (50) takes a form

$$\psi_+ = \mu\overline{\omega} + U_1(z) - U(z) + B_1\omega - \frac{1}{4\pi it}\int\limits_{|t|=1} B_1\omega dt.$$

Return from B_1 to B

$$\psi_+ = \mu\overline{\omega} + U_1 - U + B\omega - \frac{D(\omega)}{t}, \quad |t| = 1. \tag{2.51}$$

Here $D(\omega)$ denotes the functional

$$D(\omega) = \frac{1}{4\pi i}\left[\int\limits_{|t|=1} B\omega dt + \operatorname{Re}\int\limits_{|t|=1} B\omega dt\right]. \tag{2.52}$$

When we find the boundary-value problem (51) solution we may prove, as was done in Section 2.4, that the functional $c(\omega)$ in (49) vanishes on this solution. It means that we have solved problem (47) and, therefore, initial problem (45).

If problem (51) has a solution $\psi_+(t), \omega(t)$ from the class of functions $A(|t| < 1)$, then these functions have an analytic continuation through the circle $|t| = 1$ and, therefore, they satisfy the Hölder condition for $|t| = 1$. This, and the fact that $\overline{\omega(t)}$ is the limit value of $\omega_*(t)$, analytic for $|t| > 1$, $\omega_*(\infty) = 0$ yield, on the basis of the Sokhotskii-Plemel formula, the singular equation equivalent to the boundary-value problem (51)

$$\overline{\omega(t)} = \frac{1}{\mu}\left[\frac{1}{2\pi i}\int\limits_{|\tau|=1} \frac{B\omega(\tau)}{\tau - t}d\tau - \frac{1}{r}B\omega(t) + (U - U_1)(z) + \frac{D\omega}{t}\right], \quad |t| = 1. \tag{2.53}$$

The requirements of the function $rt+\omega(t)$ univalence lead to the necessity of evaluation of $\omega'(t)$ for $|t| = 1$. Therefore, it is natural to solve (53) in the space of the functions $C^1_\alpha(|t| = 1)$, where α is an arbitrary number from the interval $(0, 1)$. But, because of the complexity of calculating the functional and operator norms, we shall introduce another way of solving (53) or (51).

3^0. The essence of the suggested method for solving (51) is the analytic continuation of the boundary condition (51) from the circle $|t| = 1$ to the circle $|t| = \rho$, where ρ is a certain number, $\rho < 1$. Let us find this ρ.

The function $U(rt)$ is analytic for $|rt| \geq R$ or for $|t| \geq R/r$. Let us find the analyticity domain of $U[rt + \omega(t)]$. Assume that

$$\max|\omega(t)| \leq \frac{r-R}{\gamma}; \tag{2.54}$$

$\gamma \geq 2$ we shall determine further. Then for $|t| = 1$ we have

$$|rt + \omega(t)| \geq r - \frac{r-R}{\gamma} = R + \frac{\gamma-1}{\gamma}(r-R). \tag{2.55}$$

Thus, if (54) holds, the function $U[rt + \omega(t)]$ is determined for $|t| = 1$. Now we may continue analytically this function from $|t| = 1$ to $|t| = \rho$,

$$\rho = \frac{1}{2}\left(1 + \frac{R}{r}\right) = \frac{r+R}{2r}. \tag{2.56}$$

Really, for such ρ we have

$$|rt| = \frac{r+R}{r} = R + \frac{r-R}{2},$$

due to (54). By means of the principle of the analytic function modulus we obtain

$$|rt + \omega(t)| \geq R, \quad |t| = \rho,$$

i.e., we get into the domain of definition. Really, we have the more exact inequality which is obtained by means of the Schwartz lemma for analytic in the disk $|t| < 1$ function $\omega(t)$, $\omega(0) = 0$ with condition (54)

$$|\omega(t)| \leq \frac{r-R}{\gamma} \cdot \rho, \quad |t| = \rho, \tag{2.57}$$

$$|rt + \omega(t)| \geq \rho r - \rho\frac{r-R}{\gamma} = \rho\left[\frac{(\gamma-1)r + R}{\gamma}\right]. \tag{2.58}$$

We shall write (51) in the form

$$\psi_+(t) = \mu\omega_*(t) + T(t), \quad |t| = 1, \tag{2.59}$$

$$T(t) = U_1(rt) - U(rt) + B\omega(t) - \frac{D(\omega)}{t}.$$

As follows from above, the function $T(t)$ is analytic in the ring $\rho \le |t| \le 1$. Formula (59) gives the analytic continuation of the function $\omega_*(t)$ from $|t| > 1$ into $|t| \ge \rho$, i.e.,

$$\psi_+(t) = \mu\omega_*(t) + T(t), \quad |t| = \rho. \tag{2.60}$$

Conversely, (60) gives an analytic continuation of the function $\psi_+(t)$ from $|t| \le \rho$ into $|t| \le 1$. Now, the Cauchy formula gives

$$\overline{\omega(t)} = \frac{1}{\mu}\left[U[z] - U_1[z] + \frac{D(\omega)}{t} - \frac{1}{2\pi i}\int\limits_{|\tau|=\rho} \frac{B\omega(\tau)}{\tau - t}d\tau\right], \quad |t| = 1. \tag{2.61}$$

This is the base equation for the following research. In contrast to (53) it has no singular operator, and it allows us to use for its solution the uniform norm

$$\|\cdot\| = \max_{|t|=1}|\cdot|.$$

So, the result of our research is as follows:

LEMMA 2.3. Boundary-value problem (51) in the class of the functions $A(|t| < 1)$ is equivalent to equation (61) under condition (54).

We may write (61) in the form

$$\omega = \Gamma\omega, \tag{2.62}$$

where $\Gamma\omega$ is the value conjugated to the right side of (61). We shall solve (62) in the Banach space $A(|t| < 1)$ satisfying (44). By the construction, operator Γ is determined in the ball (54) of this space.

4^0. Now we shall show that a certain ball $\|\omega\| \le d$ lying in the ball (54) operator Γ maps into itself, and therefore, Γ is a contracting operator. To determine d we prove the following lemma.

LEMMA 2.4. In the ball $\|\omega\| \le d$, for $(r/R) \le (3/5)$ the following estimate holds

$$\|\Gamma\omega\| \le ad^2 + bd + c, \tag{2.63}$$

$$c = \frac{\lambda R^3}{c_0 r}, \quad b = \frac{9\lambda_1 R^4}{c_0 r^2}, \quad a = \frac{a_1(\alpha,\gamma)}{r} + \frac{a_2(\gamma)\lambda_2 R^5}{c_0 r^3},$$
$$a_1 = \left(\frac{\gamma}{\gamma - 1 + \alpha}\right)^3 \frac{3-\gamma}{1-\gamma}, \quad a_2 = \frac{9}{2}\left(\frac{\gamma}{\gamma - 1}\right)^5, \quad \alpha = \frac{R}{r}, \tag{2.64}$$

where $\lambda, \lambda_1, \lambda_2$ are determined in (42), (43).

PROOF. From (62), (61), due to designations (52), we have

$$\|\Gamma\omega\| \le \frac{1}{\mu}\left[\|U(z) - U_1(z)\| + \|B\omega\| + \frac{\rho}{1-\rho}\max_{|t|=\rho}|B\omega(t)|\right]. \tag{2.65}$$

Using (42) and the Schwartz lemma evaluate the quantity

$$\|U[z(t)] - U_1[z(t)]\| = \max_{|z|=r}|U(z) - U_1(z)|.$$

To this purpose we introduce the auxiliary function

$$g(\zeta) = U\left(\frac{1}{\zeta}\right) - U_1\left(\frac{1}{\zeta}\right), \quad \zeta = \frac{1}{z}$$

analytic for $|\zeta| \leq 1/R$ and, due to (41), having a zero of the third order in a zero. Therefore, the function $g(\zeta)/\zeta^2$ vanishes for $\zeta = 0$, and, due to (42)

$$\left|\frac{g(\zeta)}{\zeta^2}\right| \leq \lambda \left|\zeta^2\right| = \lambda R^2.$$

Then, by the Schwartz lemma

$$\left|\frac{g(\zeta)}{\zeta^2}\right| \leq \frac{\lambda R^3}{r}, \quad |\zeta| = \frac{1}{r},$$

hence

$$|U(z) - U_1(z)| \leq \frac{\lambda_0 R^3}{r^3}, \quad |z| = r. \tag{2.66}$$

Analogously we obtain

$$\left|(U(z) - U_1(z))'\right| \leq \frac{\lambda_1 R^4}{r^4}, \quad |z| = r. \tag{2.67}$$

Further, we use the well known inequality

$$\|U(z) + U'(z)\omega - U(z+\omega)\| \leq \frac{M}{2}\|\omega\|^2, \tag{2.68}$$

where M is the maximum of the function $|U'(\xi)|$ for ξ from the domain of definition of $z + \omega$. In our case (54) implies (55) and, therefore, by the maximum principle

$$M = \max|U''(\xi)| \quad \text{for } |\xi| = \frac{(\gamma-1)r + R}{\gamma}. \tag{2.69}$$

Since

$$|U''(\zeta)| = |U_1'' - (U_1 - U)''| \leq |U_1''| + |(U_1 - U)''|,$$

then evaluating the second term by means of (43) and the Schwartz lemma, we obtain for $|\zeta| = [(\gamma-1)r + R]/\gamma$

$$M \leq 2c_0\left(\frac{\gamma}{(\gamma-1)r+R}\right)^3 + \lambda_2\left(\frac{\gamma R}{(\gamma-1)r+R}\right)^5. \tag{2.70}$$

The first two terms in (65), on the basis of (66)–(70), we may estimate as follows:

$$\frac{1}{\mu}\left(\|U - U_1\| + \|B\omega\|\right) \leq \frac{1}{\mu}\left\{\frac{\lambda_0 R^3}{r^3} + \frac{\lambda_1 R^4}{r^4}\|\omega\| + \right.$$

$$\left. + \left[c_0\left(\frac{\gamma}{(\gamma-1)r+R}\right)^3 + \frac{\lambda_2}{2}\left(\frac{\gamma R}{(\gamma-1)r+R}\right)^5\right]\|\omega\|^2\right\}. \tag{2.71}$$

Now, evaluate the latter term in (65)

$$\frac{\rho}{\mu(1-\rho)}\max_{|t|=\rho}|B\omega(t)| \le \frac{\rho}{\mu(1-\rho)}\max_{|z|=r\rho}\left|(U(z)-U_1(z))'\right| + \max_{|t=\rho|}|\omega(t)| +$$
$$+\max_{|t|=\rho}|U(z)-U'(z)\omega - U(z+\omega)|. \tag{2.72}$$

We have

$$\max\left|(U(z)-U_1(z))'\right| \le \lambda_1\left(\frac{rR}{r+R}\right)^4.$$

Here we again apply the Scwartz lemma. The second term in (72) may be evaluated by means of (68), where, on the basis of (58)

$$M = \max|U''(\zeta)| \quad \text{for} \quad |\zeta| = \rho\frac{(\gamma-1)r+R}{\gamma}.$$

Therefore, proceeding as when proving (70), we obtain

$$M \le 2c_0\left[\frac{\gamma}{\rho[(\gamma-1)r+R]}\right]^3 + \lambda_2\left[\frac{\gamma R}{\rho[(\gamma-1)r+R]}\right]^5.$$

Combining the latter estimates and taking into account the inequality

$$\max_{|t|=\rho}|\omega(t)| \le \rho\,\|\omega\|,$$

we can show that (72) takes a form

$$\frac{\rho}{\mu(1-\rho)}\max_{|t|=\rho}|B\omega| \le \frac{\rho}{\mu(1-\rho)}\left\{\lambda_1\rho\left(\frac{2R}{r+R}\right)^4\|\omega\| + \right.$$
$$\left. + \rho^2\|\omega\|^2\left(c_0\left[\frac{\gamma}{\rho[(\gamma-1)r+R]}\right]^3 + \frac{\lambda_2}{2}\left[\frac{\gamma R}{\rho[(\gamma-1)r+R]}\right]^5\right)\right\}. \tag{2.73}$$

5^0. Continuation of the lemma proof.

On the basis of (71), (73), the equality $\mu r^2 = c_0$, and the designation $R/r = \alpha$, the total coefficient at $\|\omega\|$ is

$$\frac{\lambda_1 R^4}{c_0 r^2}(1+8f(\alpha)), \quad f(\alpha) = \frac{1}{(1+\alpha)^2(1-\alpha)}.$$

The unique extremal point of the function $f(\alpha)$ in the interval $(0,1)$ is the minimum point $\alpha = 1/3$. It is easy to show that for

$$\alpha \le R/r \le 3/5 \tag{2.74}$$

we have $f(\alpha) \le 1$. Thus the coefficient b in (63), (64) is obtained. The sum of the latter terms (72), (73) gives the value

$$\frac{\lambda_2 R^5}{c_0 r^3}\frac{1}{2}\left(\frac{\gamma}{\gamma-1+\alpha}\right)^5(1+8f(\alpha)),$$

which, under the condition (74), is majorized by $a_2(\gamma)$ in (64). Finally, combining the third term in (71) and the second in (73) we obtain the quantity $a_1(\alpha,\gamma)$ which is in (64). Hence, the lemma is proved.

The critical points of $a_1(\alpha,\gamma)$ (γ is a parameter) are defined by the formula

$$\alpha_{1,2} = \frac{7 \pm \sqrt{16+6\gamma}}{3}.$$

In particular, for $\gamma \le 5.5$ in the interval $(0,\alpha_*)$, $\alpha_* \le 3/5$ there are no critical points and the following relation holds:

$$a_1(\alpha,\gamma) \le a_1(\alpha_*,\gamma) \le a_1(3/5,\gamma) = 6\left(\frac{\gamma}{\gamma-0.4}\right)^3. \tag{2.75}$$

6^0. Equating the right side of (63) to d we obtain the quadratic equation (here we follow the Newton method of solving nonlinear equations (Kantorovich and Akilov, 1977)), the least root of which

$$d_1 = \frac{1-b-\sqrt{(1-b)^2-4ac}}{2a} \tag{2.76}$$

exists if

$$b<1, \quad (1-b)^2-4ac \ge 0. \tag{2.77}$$

Note that inequalities (77) hold for r being sufficiently large, since a, b, c tend to zero for $r \to \infty$.

So, if (77) holds, then Γ maps the ball $\|\omega\| \le d_1$ onto itself.

When confirming the compressibility of Γ in the ball $\|\omega\| \le d_1$, we obtain the inequality

$$\|A\omega_1 - A\omega_2\| \le b + 2ad_1\,\|\omega_1-\omega_2\|,$$

and on the basis of (76)

$$b+2ad_1 = 1-\sqrt{(1-b_1)^2-4ac} < 1.$$

Thus, our problem is reduced to solving, relative to r, numerical inequalities (77) with the conditions

$$\frac{r}{R} \le 3/5, \quad d_1 = \frac{r-R}{\gamma}. \tag{2.78}$$

Besides, we have to establish the univalence of the obtained solution.

7^0. Now we shall find such a number r_1, that for $r \ge r_1$ the second inequality in (77) holds. This inequality may be reduced to the quadratic inequality by means of the change $c_0 r^2 = x$:

$$x^2 - (18\lambda_1 R^4 + 4a_1\lambda R^3) + (81\lambda_1^2 - 4a_2\lambda\lambda_2)R^8 \ge 0.$$

This holds for $x \ge x_1$, where

$$x_1 = 2a_1\lambda R^3 + 9\lambda_1 R^4 + 2R^3(a_1^2\lambda^2 + 9a_1\lambda\lambda_1 R + a_2\lambda\lambda_2 R^2)^{1/2}. \tag{2.79}$$

So, for $c_0 r^2 \geq x_1$, or

$$r \geq r_1, \quad r_1 = \sqrt{\frac{x_1}{c_0}} \tag{2.80}$$

the second inequality in (77) holds. The first inequality in (77), being of the form $c_0 r^2 \geq 9\lambda_1 R^4$, holds all the more, since it immediately follows from (79).

By the number r_1 we find $d_1(r_1) = d_*$

$$d_* = \frac{1-b}{2a} = \sqrt{\frac{c}{a}}.$$

Taking the designations (64) we obtain

$$d_* = \left(\frac{\lambda R^3}{a_1 c_0 + \dfrac{a_2 \lambda_2 R^5}{r_1^2}} \right)^{1/2} \leq \left(\frac{\lambda R^3}{a_1 c_0} \right)^{1/2}. \tag{2.81}$$

From (79) it follows that $4a_1 \lambda R^3 \leq c_0 r_1^2$; therefore

$$d_* \leq \frac{r_1}{2a_1}. \tag{2.82}$$

The ball $\|\omega\| \leq d_*$ lies in the ball $\|\omega\| \leq (r-R)\gamma$ if

$$\frac{r_1}{2a_1} \leq \frac{r_1 - R}{\gamma}$$

holds. Let $(R/r_1) \leq d_*$, then we obtain the inequality

$$\gamma \leq 2a_1(1-\alpha_*), \tag{2.83}$$

which determines γ. In particular, for $\alpha_* = 3/5$, $\gamma = 5.65$ the formula in (64) gives $a_1 = 7.5$, and (83) holds. From (64) it follows that $a_2 = 12$ and x_1 determined by (79) is the same as x_1 from Theorem 2.5.

8^0. Now we shall show that $rt + \omega(t)$ is univalent, where $\omega(t)$ solves (62), $\|\omega\| \leq d_*$ and $r \geq r_1$. To this purpose we shall use the sufficient univalence condition (Avkhadiev and Aksent'ev, 1975) in the form

$$\operatorname{Re}(rt + \omega(t))' \geq 0, \quad |t| = 1.$$

This condition holds if

$$\|\omega'\| \leq r \tag{2.84}$$

holds. Now we shall prove this inequality.

Differentiating (62) with respect to t we find $\omega'(t)$, whence

$$\|\omega'\| \leq \frac{1}{\mu} \left(\max_{\bar{z}=r} |(U-U_1)'|\, r + \|B\omega\| + \frac{\rho}{(1-\rho)^2} \max_{|t|=\rho} |B\omega| \right). \tag{2.85}$$

Here, all the quantities have been evaluated before, but we shall evaluate the first term in another way than in (67). By the Cauchy formula we have

$$z^2(U(z)-U_1(z)) = -\frac{1}{2\pi i}\int\limits_{|z|=R}\frac{\zeta^2(U(\zeta)-U_1(\zeta))}{\zeta - z}d\zeta, \quad |z|>R.$$

Divide this equation by z^2 and differentiate. Then for $|z|=r$, $\alpha=(R/r)>\gamma_*$ we obtain

$$\left|(U(z)-U_1(z))'\right| \le \frac{R^3\lambda_0}{r^4}\left(\frac{2}{1-\alpha_*}+\frac{1}{(1-\alpha_*)^2}\right).$$

Now, from (85) it follows that

$$\|\omega'\| \le \left[\frac{2}{1-\alpha_*}+\frac{1}{(1-\alpha_*)^2}\right]\frac{1}{\mu}\left[\frac{R^3\lambda_0}{r^3}+\|B\omega\|+\frac{\rho}{1-\rho}\max_{|t|=\rho}|B\omega|\right].$$

The term in the second square brackets with the multiplier $1/\mu$ is less than d_*. On the basis of (63) and 6^0 we obtain

$$\|\omega'\| \le \left(\frac{2}{1-\alpha_*}+\frac{1}{(1-\alpha_*)^2}\right)d_*.$$

On the basis of (62), estimate (64) follows from

$$\left(\frac{2}{1-\alpha_*}+\frac{1}{(1-\alpha_*)^2}\right)\frac{1}{2a_1} \le 1. \tag{2.86}$$

For $\alpha_*=3/5$, $a_1=7.5$ inequality (86) holds. This completes the Theorem 2.5 proof.

9^0. The proof of Theorem 2.6 will be similar, only the derivatives of $U(z)-U_1(z)$ we evaluate not over λ_1,λ_2, but over λ by means of the Cauchy integral; as a result, the desired number μ_* will depend only on R,λ.

LEMMA 2.5. In the ball $\|\omega\|\le d$, for $(r/R)\le 1/3$, the estimate (63) holds, where

$$c=\frac{\lambda R^3}{c_0 r}, \quad b=\frac{b_1(\alpha)\lambda R^3}{c_0 r^2}, \quad a=\frac{a_1}{r}+\frac{a_2(\alpha)\lambda R^3}{c_0 r^3}, \tag{2.87}$$

$$b_1=\frac{2}{1-\alpha}+\frac{1}{(1-\alpha_*)^2}+\frac{16}{(1+\alpha)(1-\alpha)^2}+\frac{8}{(1-\alpha)^3}, \quad a_1=5.5, \tag{2.88}$$

$$a_2=\left(\frac{7}{6}\right)^5\left[3q_1+2q_1^2+q_1^3+8(3q_2+2q_2^2+q_2^3)\right], \tag{2.89}$$

$$q_1=\frac{6+\alpha}{6-6\alpha}, \quad q_2=\frac{6+\alpha^2+7\alpha}{6+\alpha^2-7\alpha}. \tag{2.90}$$

PROOF. Differentiating twice successively the Cauchy formula

$$U(z)-U_1(z)=-\frac{1}{z^2}\frac{1}{2\pi i}\int\limits_{|\zeta|=R}\frac{\zeta^2(U(\zeta)-U_1(\zeta))}{\zeta-z}d\zeta, \quad |z|>R,$$

we obtain

$$\left|(U-U_1)'\right| \le \frac{\lambda R^3}{|z|^4}\left[\frac{2\,|z|}{|z|-R}+\left(\frac{|z|}{|z|-R}\right)^2\right], \tag{2.91}$$

$$\left|(U-U_1)''\right| \le \frac{\lambda R^3}{|z|^5}\left[\frac{6\,|z|}{|z|-R}+4\left(\frac{|z|}{|z|-R}\right)^2+2\left(\frac{|z|}{|z|-R}\right)^3\right]. \tag{2.92}$$

Using (91) we obtain

$$\left|(U-U_1)'\right| \le \frac{\lambda R^3}{r^4}\left[\frac{2r}{r-R}+\left(\frac{r}{r-R}\right)^2\right], \quad |z|=r, \tag{2.93}$$

$$\left|(U-U_1)'\right| \le \frac{\lambda R^3}{r^4}\left(\frac{2r}{r+R}\right)^4\left[2\frac{r+R}{r-R}+\left(\frac{r+R}{r-R}\right)^2\right], \quad |z|=\rho r. \tag{2.94}$$

Multiplying the right side of (94) by $\rho^2/(1-\rho)$, using (65), inequalities $\max|\omega(t)| \le \rho\,\|\omega\|$, and combining it with (93) we obtain coefficient b in (87), (88). Note that b_1 is monotonic

$$b_1(\alpha) \le b_1(\alpha_*), \quad 0<\alpha\le\alpha_*<1. \tag{2.95}$$

Now let us obtain a in (87). The coefficient at $1/r$ does not depend on λ_1, λ_2; therefore, it is the same as in (64). If $\alpha \le \alpha_* = 1/3$, then, due to (75)

$$a_1(\alpha,\gamma) \le 4\left(\frac{\gamma}{\gamma-2/3}\right)^3.$$

The number γ we shall take so that (75), (83), (86) hold. Take $\gamma = 7$ then $a_1 = 5.5$. It remains to obtain a_2 in (87).

Since

$$\frac{|z|}{|z|-R} = q_1(\alpha) \quad \text{for} \quad |z| = \frac{6r+R}{7}$$

then (92) yields

$$\left|(U-U_1)''\right| \le \frac{\lambda R^3}{r^5}\left(\frac{7}{6}\right)^5\left[6q_1+4q_1^2+2q_1^3\right]. \tag{2.96}$$

Analogously, for $|z| = \rho(6r+R)/7$ we have

$$\left|(U-U_1)''\right| \le \frac{\lambda R^3}{r^5}\left(\frac{7}{\rho(6r+R)}\right)^5\left[6q_2+4q_2^2+q_2^3\right]. \tag{2.97}$$

Hence

$$\frac{\rho^3}{1-\rho}\left|(U-U_1)''\right| \le \frac{\lambda R^3}{r^5}\left(\frac{7}{6}\right)^5 8\left[6q_2+4q_2^2+2q_2^3\right].$$

Combining it with (96) and taking into account the multiplier 1/2 in (68) we obtain the coefficient $a_2(\alpha)$ in (87), (89). The lemma is proved.

The number x_1 from formula (79) is equal to

$$x_1 = \lambda R^3\left[2a_1+b_1+2(a_1^2+a_1b_1+a_2)^{1/2}\right]. \tag{2.98}$$

For $\alpha_* = 1/3$, $\gamma = 7$ we have $a_1 = 5.5$, $b_1 = 60$, $a_2 = 510$ and formula (98) gives $x_1 = 130\lambda R^3$ which coincides with the value x_1 in Theorem 2.6. The theorem is proved.

10^0. We have obtained more than what is asserted in Theorems 2.5 and 2.6. Due to (75), the diminuation of α_* leads to the diminuation of a_1 in (64) and, therefore, of x_1 in (79). First $\alpha_* = 3/5$ and after applying Theorem 2.5 we may diminuate α_* if $r_* > (5/3)R$ and, thus, we obtain the more exact estimate. The parameter γ may be diminished, but in such a way as to save inequalities (83), (86). This will make the result more precise.

In the case of example 1, assuming $\alpha_* = 1/8$, $\gamma = 8$, we obtain $a_1 = 4.65$, $a_2 = 8.77$, which gives $\mu_* = 0.0147$.

The process described for Theorem 2.5 refinement may be formulated as follows.

REMARK 2.12. Under the Theorem 2.5 conditions the number r_* may be determined as

$$r_* = \min_{0\le\alpha_*\le 3/5}\left[\max\left(\frac{R}{\alpha_*}, \sqrt{\frac{x_1(\alpha_*)}{c_0}}\right)\right],$$

where $x_1(\alpha_*)$ is given by (79).

REMARK 2.13. An *a priori* estimate from above for α follows from the condition $b < 1$, which is necessary, as it follows from (77), for compressibility of Γ in Theorem 2.5. Really, this inequality is the following

$$\alpha < \alpha_0, \quad \alpha_0 = \sqrt{\frac{c_0}{9\lambda_1 R^2}}.$$

The more exact estimate from above for the parameter α and for the number r_1 from below may be obtained from (79). Namely, due to the monotonicity of $x_1(\alpha)$, we have

$$c_0 r_1^2 \ge x_1(0),$$

$$x_1(0) = 8\lambda R^3 + 9\lambda_1 R^4 + 2R^3(16\lambda^2 + 36\lambda\lambda_1 R + 8\lambda\lambda_2 R^2)^{1/2},$$

whence

$$\alpha \le \alpha_1, \quad \lambda_1 = \sqrt{\frac{c_0 R^2}{x_1(0)}}.$$

Specifically α_1 has to be used for determining a_1 and, therefore, x_1.

REMARK 2.14. Under the conditions of Theorem 2.6 the number r_* may be determined by the formula

$$r_* = \min_{0\le\alpha_*\le 1/5}\left[\max\left(\frac{R}{\alpha_*}, \sqrt{\frac{x_1(\alpha_*)}{c_0}}\right)\right],$$

where the value $x_1(\alpha_*)$ is given by (98). In particular, for $\alpha_* = 1/8$

$$r_* = \max(8R, \sqrt{\frac{x_1}{c_0}}), \quad x_1 = 81\lambda R^3.$$

REMARK 2.15. From (98) it follows, that

$$c_0 r_*^2 \ge x_1(0) 64\lambda R^3.$$

Hence an *a priori* estimate from below for r_1 and from above for α follows

$$r_1 \geq 8\sqrt{\frac{\lambda R^3}{c_0}}, \quad \alpha \leq \frac{1}{8}\sqrt{\frac{c_0}{\lambda R}} = \alpha_*.$$

The scheme used in Theorem 2.5 does not allow us to obtain r_* less than the right side of the first inequality.

REMARK 2.16. In the suggested method of finding sufficient solvability conditions the inverse problem solution was looked for 'near' the circle $|z| = r$. We may seek a solution 'near' the domain $z(t) = r(t + a_3 t^3)$, $|a_3| \leq 1/3$, where the parameters r, a_3, μ are chosen so that the potential of the domain specified by the map $z(t)$ with the density μ will be equal to

$$U_1(z) = \frac{c_0}{z} + \frac{c_2}{z^3}.$$

Here, c_0 and c_2 on the basis of Section 1.3, are the following

$$c_0 = \mu r^2(1 + 3\,|a_3|^2), \quad c_2 = \mu r^4 \overline{a_3}.$$

The linear problem in this case is solved explicitly and the function $U(z) - U_1(z)$ decreases more rapidly than earlier. This allows us to obtain more exact estimates of the type $\mu \leq \mu_*$ which guarantee the existence of the inverse problem solution.